

Measuring Socio-Ethical Engagement with Generative AI: Scale Development via Exploratory Factor Analysis

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ABSTRACT

As the use of generative artificial intelligence (GenAI) systems grows in daily life, there is a need to assess how users interact with these tools in socially and ethically informed ways. This study introduces a multidimensional scale to measure socio-ethical AI engagement competencies, reflecting users' ability to evaluate, interpret, and ethically use AI-generated content, and to critically consider its broader social impacts and power dynamics. Drawing from interdisciplinary frameworks in AI literacy, self-efficacy, and AI ethics, we constructed an initial item pool related to social and ethical engagement.

Responses from 200 participants to an 18-item instrument were analyzed using Exploratory Factor Analysis (EFA) to capture four dimensions: critical appraisal, critical comprehension, ethical behavior, and anthropomorphic interaction. The scale demonstrated strong internal consistency. This work contributes to a theoretically grounded and empirically supported instrument for assessing critical and ethical engagement with GenAI, which has implications for AI literacy, responsible technology use, and curriculum design.

KEYWORDS

AI Literacy, AI competencies, Scale development, Generative AI, Socio-ethical engagement

INTRODUCTION

Artificial Intelligence (AI) has established a powerful presence across diverse sectors, such as healthcare, finance, education, and entertainment, enabling machines to perform tasks traditionally associated with human intelligence, encompassing visual perception, speech recognition, decision-making, and language understanding (Almatrafi et al., 2024; Leander & Burriss, 2020; Russell & Norvig, 2016). Generative AI (GenAI) has achieved widespread adoption and accessibility through its capacity to generate content across multiple modalities, including audio, images, text, simulations, and videos (Chiu, 2024; IBM, n.d.). For instance, OpenAI's ChatGPT, released in November 2022, accumulated over 100 million users within two months, becoming one of the most rapidly adopted GenAI-based services of all time (Henny & Pennock, 2023). GenAI's widespread utilization has changed the socio-technical landscape, which leads some scholars to claim it signals a paradigm shift across different societal spheres (Leander & Burriss, 2020; Southworth et al., 2023; Vinchon et al., 2023). However, this rapid adoption has also sparked significant controversy that raises major concerns about its ethical issues, potential risks, and true benefit to society (Crawford, 2021).

As GenAI is more embedded in daily life and formal settings, understanding its risks, limitations, and socio-ethical implications has become increasingly important (Giustini & Dastyar, 2024). Despite its advanced capabilities, many users remain unaware of the underlying technology and ethical concerns (Ng et al., 2021). This emphasizes the importance of AI literacy and competencies related to social and ethical engagement, which equip individuals to analyze, critique, and process the information they encounter. Indeed, AI literacy is not just in understanding the technology itself but also recognizing its social and ethical implications, which are essential for gaining a deeper and more nuanced understanding of the socio-ethical aspects of various contexts, such as education, work, and daily life (Kandlhofer et al., 2016; Long & Magerko, 2020; Ng et al., 2021; Veldhuis et al., 2025).

In addressing AI literacy needs, several scales have been developed, capturing diverse dimensions such as affective, behavioral, cognitive, and ethical aspects (Ma & Chen, 2024; Ng et al., 2023; Soto-Sanfiel et al., 2024; Wang et al., 2023). For instance, Wang et al.'s (2023) AI Literacy Scale and Ng et al.'s (2023) questionnaire include dimensions like awareness, evaluation, and ethics. As Bozkurt (2024) emphasizes, navigating GenAI requires comprehensive literacy encompassing theoretical knowledge (know *what*), practical application (know *how*), and an in-depth ethical and philosophical understanding of its implications (know *why*). However, fewer instruments have focused specifically on how users engage with the social and ethical dimensions of GenAI, such as accountability and awareness of ascribing human-like interaction. These competencies are especially important as GenAI becomes more embedded in decision-making, where the technology can exert influence and agency over high-stakes outcomes for individuals, often with limited accountability. Assessing this engagement is crucial for fostering

responsible and critical interaction with GenAI. Building on these frameworks, this study aims to contribute to the GenAI literature by developing a scale that assesses social and ethical engagement for GenAI.

LITERATURE REVIEW

Defining Socio-Ethical Engagement of Generative AI

Socio-ethical engagement with GenAI refers to how individuals interpret, evaluate, and respond to GenAI systems in ways that are informed by social, ethical, and contextual awareness. Foundational work on critical literacy, particularly Freire's (1970) early contributions, has shaped this space. His definitions are often summarized as "the ability to analyze, critique, and transform social, cultural, and political texts and contexts" (Veldhuis et al., 2025, p. 2), and have influenced the development of digital and AI literacies. Building on this foundation, recent scholarships have proposed more context-specific forms of engagement with AI. For example, Bali (2023) provided a holistic view of AI literacy that goes beyond understanding technical aspects to include awareness of biases, ethics, social inequalities, prompt engineering, and the contexts in which AI is beneficial. This holistic approach aims to empower individuals to effectively navigate various AI use cases.

Similarly, Veldhuis et al. (2025) defined AI literacy as "the capacity to analyze, critique and transform AI's critical implications (p. 14)." Long and Magerko (2020) highlighted the critical dimension of AI literacy, defining it as "a set of competencies that enables individuals to critically evaluate AI technologies, communicate and collaborate effectively with AI, and use AI as a tool online, at home, and in the workplace." In addition, the concept of critical literacy is further defined as the ability to critically assess all core areas, such as AI models, tool outputs, usage needs, and the impact through various critical lenses, including social and power dynamics, and ethical implications (Ball & Huang, 2023; Carolus et al., 2023; Huang & Ball, 2024; Laupichler, Aster, & Raupach, 2023; Laupichler et al., 2022; Picasso et al., 2024; Register & Ko, 2020).

Together, these perspectives inform our conceptualization of socio-ethical GenAI engagement as a multidimensional construct. Specifically, we define it as the capacity to critically evaluate the credibility and risks of AI-generated content, interpret that content within relevant personal and social contexts, evaluate the potential outcomes and fairness of using such content, and recognize how it may influence trust and decision-making. These dimensions guided our scale development, allowing us to operationalize this form of engagement through items that reflect everyday interactions with GenAI systems.

Justification for Dimensions

Building on the literature reviewed, we conceptualize socio-ethical engagement with GenAI as a multidimensional construct that reflects users' evaluative, interpretive, ethical, and perceptual interactions with GenAI systems. Specifically, we identify four distinct but complementary dimensions that together describe how individuals make meaning from, act upon, and respond to GenAI outputs in socially and ethically informed ways: Critical appraisal, critical comprehension, ethical behavior, and anthropomorphic interaction.

Critical appraisal involves systematic evaluation of the trustworthiness, value, and relevance of AI-generated content (Laupichler, Aster, & Raupach, 2023; Mhaskar et al., 2009). As GenAI raises challenges in examining credibility (Ou et al., 2024), bias and representativeness of the data used in AI models (Mikalef & Gupta, 2021; Qian et al., 2024), and hallucinations (Bearman et al., 2024) can be identified as key concerns. *Critical comprehension* refers to the user's skill in verifying and interpreting AI-generated information (Hwang et al., 2023), which emphasizes the importance of reflective interpretation in various contexts, a challenge amplified by the often opaque "black box" nature of GenAI systems and their potential to oversimplify complex societal issues (Gautam et al., 2024). *Ethical behavior* involves critically evaluate information acquired through digital tools and technologies from an ethical and moral perspective (Hwang et al., 2023). *Anthropomorphic interaction* assesses how users perceive GenAI systems as human-like (Wang & Chuang, 2024), a tendency that critical scholarship warns can obscure the underlying statistical processes and human decisions. This can make it harder for users to critically understand, evaluate, and contextualize AI outputs, but can also promotes "AI Realism", the belief that AI is an inevitable force (Gautam, 2024; Gebru & Torres, 2024). Conversely, research (Markus et al., 2024; Pinski & Benlian, 2023) show that higher AI literacy helps users to mitigate this tendency and more critically engagement with these systems

METHODOLOGY

Scale Development

The scale development process for measuring socio-ethical engagement for GenAI followed established guidelines from existing literature (Carpenter, 2018). First, we conducted an extensive literature review to identify previously validated scales and frameworks aligned with our socio-ethical perspective, which specifically focused on critical literacy, ethical behavior, critical comprehension, and human-AI interactions. We initially selected and adapted items from prior work relevant to social and ethical dimensions of AI literacy (Hwang et al., 2023; Karaca et al., 2021; Laupichler, Aster, et al., 2023; Laupichler, Aster, & Raupach, 2023; Lee & Park, 2024; Ng et al., 2023; Pinski & Benlian, 2023; Wang & Chuang, 2024). After developing an initial item pool, we carefully reviewed the items to

ensure conceptual clarity, relevance, and coherence. Based on these evaluations, items were refined and finalized into a structured questionnaire that comprised four dimensions: critical appraisal, ethical behavior, critical comprehension, and anthropomorphic interaction. Each dimension consisted of four to ten items and was measured on a 7-point Likert scale.

Data Collection and Sample

The final questionnaire included the developed critical literacy scale and demographic questions, including age, gender, ethnicity, and education level. Data were collected online via the CloudResearch Connect platform by recruiting 200 participants located in the United States, who self-reported their responses to the critical literacy scale items. Participants were instructed to consider their general interactions with various GenAI tools (e.g., chatbots, image generators) when responding. Among all the participants, 41% of the participants are females. Ages ranged from 18 to 65 and above, with the largest age group being 35-44 years (34%) and 25-34 years (32%). Most identified as White or Caucasian (74.5%), followed by Black/African American (14.5%), Asian (10%). Educational backgrounds included bachelor's degree (50.5%), some college experience but no degree (17.5%), graduate or professional degrees (17.5%), associate/technical degrees (8%), and high school diploma or GED (6%).

Data Analysis

Data were analyzed using SPSS 30.0. We first screened for data quality by checking for incomplete responses. Next, we tested the appropriateness of the data for exploratory factor analysis (EFA) using the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy and Bartlett's test of sphericity. Following standard recommendations by Kline's (1994) and Hoogland and Boomsma's (1998), we confirmed that our sample size of 200 was appropriate for conducting a robust EFA. We conducted a principal component analysis (PCA) with Varimax rotation to identify underlying factors. Factors were retained based on eigenvalues greater than 1.0, and items were kept if they showed factor loadings of .60 or higher with cross-loadings above .40. Items with communalities below .40 were removed. After performing the initial EFA, items not meeting these criteria were deleted. The final EFA resulted in an 18-item scale organized into four clearly interpretable factors. Reliability analysis was then conducted using Cronbach's alpha to assess internal consistency for the overall scale and individual subscales.

RESULTS

Reliability Analysis

To measure GenAI literacy, we initially selected 24 questions based on previous literature related to GenAI literacy studies. We performed a reliability analysis for each item using Cronbach's alpha. The scale demonstrated high internal consistency, with a Cronbach's alpha of .889, which indicates that the items are highly consistent and reliably measure the intended construct. Cronbach's alpha coefficients for each factor ranged from .731 to .933, while the reliability for the entire 18-item scale was .864. This indicates consistent measurement across both the sub-dimensions and the overall scale. According to MacCallum et al. (2001; 1999), a reliability coefficient above .70 is considered acceptable, and values above .90 are considered excellent. The high reliability in this study may be partly attributed to the number of well-performing items retained after EFA.

Exploratory Factor Analysis (EFA)

The sample size ($n = 200$) met the assumptions for exploratory factor analysis (EFA). The Kaiser-Meyer-Olkin (KMO) measure was .854, indicating adequate sampling since it exceeds the recommended threshold of .50. Bartlett's Test of Sphericity was statistically significant ($\chi^2 = 2658.990$, $df = 153$, $p < .001$), confirming that the correlation matrix is suitable for factor analysis.

Next, we performed principal component analysis using Varimax rotation. MacCallum et al. (2001; 1999) advocate that all items in a factor model should have communalities of over 0.60 or an average communality of 0.7 to justify performing a factor analysis with small sample size. Therefore, we extracted factors if they had eigenvalues greater than 1.0 (Hair et al., 2019), and items were retained if they had factor loadings of .60 or higher. Items with cross-loadings below .40 and communalities less than .40 were removed, as this indicated a low contribution to the scale (Tabachnick & Fidell, 2013). After the initial EFA, we removed items that did not meet these criteria. In the final analysis, 18 items are grouped into four factors, explaining 73% of total variance.

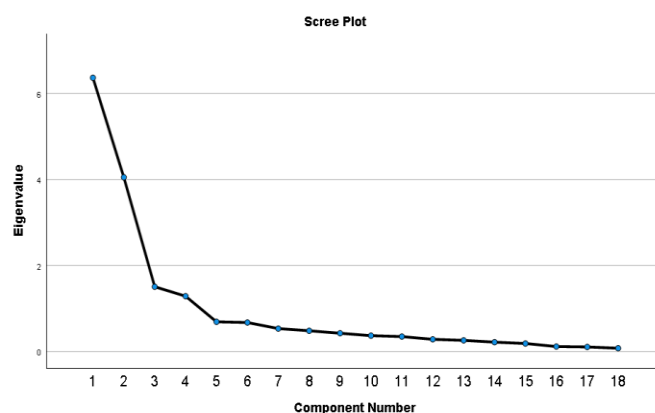


Figure 1. Scree plot of the eigenvalues for each component

Final Scale Structure

As presented in Table 1, the EFA resulted in a final scale of 18 items representing four factors. Factor 1: Critical Appraisal (8 items) measures participants' critical awareness and appraisal of ethical considerations related to GenAI. Factor 2: Anthropomorphic interaction (5 items) reflects participants' perceptions of GenAI as human-like during interactions. Factor 3: Critical comprehension (3 items) measures the participants' ability to critically evaluate and verify AI-generated content. Factor 4: Ethical behavior (2 items) relates to participants' awareness and willingness to responsibly use GenAI information, including awareness of consequences and accountability.

Factor (Cronbach's α)	Item*	Factor Loadings			
Factor 1: Critical Appraisal ($\alpha = .928$)	1. I can explain why data privacy must be considered in GenAI development and use.	.875			
	2. I can explain why data security must be considered in GenAI development and use.	.852			
	3. I can identify ethical issues surrounding GenAI.	.816			
	4. I can describe risks that may arise when using GenAI systems.	.835			
	5. I can name weaknesses of GenAI.	.713			
	6. I can describe potential legal problems that may arise when using GenAI.	.734			
	7. I can critically reflect on the potential impact of GenAI on individuals and society.	.788			
	8. I can explain why data is important in the development and application of GenAI.	.746			
Factor 2: Anthropomorphic Interaction ($\alpha = .933$)	9. Interaction with GenAI technologies/products is vivid, like chatting with a real person.		.857		
	10. I think GenAI technologies/products express content in a unique way, like a real person.		.904		
	11. The dialogue with GenAI technologies/products is no different from that with real people.		.832		
	12. I think the tone of GenAI technologies/products is the same as that of real people.		.910		
	13. GenAI technologies/products express themselves in interactive text just like real people.		.928		
Factor 3: Critical Comprehension ($\alpha = .769$)	14. When in doubt about something presented by GenAI, I check whether it is true or not.			.734	
	15. I can judge whether the content recommended by GenAI intentionally insists on one side.			.773	
	16. I am aware of how to verify that the content presented by GenAI is reliable.			.813	
Factor 4: Ethical Behavior ($\alpha = .731$)	17. When using GAI-based big data, I consider whether the information is illegal or reliable.				.827
	18. Whether I accept a GenAI recommendation, I wonder who is responsible for the outcome.				.879

*Note: The item statements in this table are shortened versions of the item statements included in the instrument

Table 1. Socio-ethical Engagement Scale for GenAI Items and Respective Factor Loadings/Cronbach Alpha

DISCUSSION

This study extends critical literacy into the GenAI context by introducing a socio-ethical scale grounded in both early foundational literacy theory (e.g., Freire, 1970) and recent AI literacy frameworks (e.g., Veldhuis et al., 2025; Bali, 2023). While prior tools focus on technical understanding or behavioral intention (Ng et al., 2023; Wang et al., 2023), our scale captures how users engage socially and ethically with GenAI outputs. The four-factor structure reflects a broader conceptual shift in AI literacy, one that goes beyond functional understanding to include the social and ethical implications of AI.

The four dimensions revealed imply that true AI literacy is a multifaceted competency requires users to practice critical appraisal of AI's credibility, critical comprehension of its contextual limits, responsible ethical behavior in using its outputs, and a critical resistance to problematic anthropomorphic interaction. Moving forward, this scale can serve as a diagnostic tool to guide future work for educators to design targeted curricula, for researchers to empirically investigate these competencies, and for designers to build more transparent and responsible AI systems.

CONCLUSION

This study introduces a socio-ethical scale for measuring users' engagement with GenAI, extending critical literacy into new technological contexts. By capturing four key dimensions (critical appraisal, comprehension, ethical behavior, and anthropomorphic interaction), the scale offers a practical tool for assessing user engagement across AI-driven environments and may provide valuable insights for future research and practice aimed at designing interventions or policies that encourage more reflective and ethical use of GenAI. While further validation is needed through larger samples and more specific populations (e.g., K-12 or college students), this study serves as a preliminary step in understanding social and ethical engagement among GenAI users. Future work should also apply methods such as confirmatory factor analysis and structural equation modeling to validate the structure of the scale. Building on this foundation, this scale can inform the design of educational initiatives that foster social and ethical engagement with GenAI technologies.

GENERATIVE AI USE

In preparing this manuscript, the author(s) used ChatGPT-4o to improve readability and language. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the submitted manuscript.

AUTHOR ATTRIBUTION

First Author: conceptualization, methodology, data curation, formal analysis, visualization, writing – original draft; Second Author, Third Author: conceptualization, methodology, writing – original draft; Fourth Author, Fifth Author: project administration, supervision, writing – review and editing.

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